4.1

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \\ 0 & 2 & 4 \end{bmatrix}$$

Sarrus Rule:

$$1.4.4 + 2.2.5 + 3.6.0 - 0.4.5 - 1.2.6 - 4.2.3$$

$$= 16 + 20 - 12 - 24 = 0$$

Laplace expansion (first row):

$$\det(A) = (-1)^{1+1} a_{11} \det(A_{1,1}) + (-1)^{2+1} a_{12} \det(A_{1,2})$$

$$+ (-1)^{3+1} a_{13} \det(A_{1,3})$$

$$= (-1)^2 . 1 . \det(A_{1,1}) + (-1)^3 . 3 . \det(A_{1,2}) + (-1)^4 . 5 \det(A_{1,3})$$

$$\det(A_{1,1}) = \begin{vmatrix} 4 & 6 \\ 2 & 4 \end{vmatrix} = 16 - 12 = 4$$

$$\det(A_{1,2}) = \begin{vmatrix} 2 & 6 \\ 0 & 4 \end{vmatrix} = 8 - 0 = 8$$

$$\det(A_{1,3}) = \begin{vmatrix} 2 & 4 \\ 0 & 2 \end{vmatrix} = 4 - 0 = 4$$

$$= 1.1.4 + (-1).3.8 + 1.5.4 = 4 - 24 + 20 = 0$$

4.2

$$\begin{bmatrix} 2 & 0 & 1 & 2 & 0 \\ 2 & -1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ -2 & 0 & 2 & -1 & 2 \\ 2 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{matrix} \\ -R_1 \\ \\ +R_1 \\ -R_1 \end{matrix} \rightsquigarrow \begin{bmatrix} 2 & 0 & 1 & 2 & 0 \\ 0 & -1 & -1 & -1 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 3 & 1 & 2 \\ 0 & 0 & -1 & -1 & 1 \end{bmatrix} \begin{matrix} \\ \\ +R_2 \\ \\ \\ \end{matrix} \rightarrow \begin{bmatrix} 2 & 0 & 1 & 2 & 0 \\ 0 & -1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 3 & 1 & 2 \\ 0 & 0 & -1 & -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 1 & 2 & 0 \\ 0 & -1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 3 & 1 & 2 \\ 0 & 0 & -1 & -1 & 1 \end{bmatrix} \begin{matrix} \\ \\ \\ -3R_3 \\ +R_3 \end{matrix} \rightsquigarrow \begin{bmatrix} 2 & 0 & 1 & 2 & 0 \\ 0 & -1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & -1 & 4 \end{bmatrix} \begin{matrix} \\ \\ \\ \\ +R_4 \end{matrix}$$

$$\downarrow$$

$$\begin{bmatrix} 2 & 0 & 1 & 2 & 0 \\ 0 & -1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & -3 \end{bmatrix}$$

Now that the matrix is in upper triangular form, we find the determinant by multiplying the diagonal elements.

$$2 \cdot (-1) \cdot 1 \cdot 1 \cdot (-3) = 6$$

4.3

$$\begin{vmatrix} 1-\lambda & 0 \\ 1 & 1-\lambda \end{vmatrix} = (1-\lambda)^2 = 0 \qquad \lambda = 1$$

plugging in $\lambda = 1$

$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad \text{we get } E_1 = \text{span}\left\{\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}$$

$$\begin{vmatrix} -2-\lambda & 2 \\ 2 & 1-\lambda \end{vmatrix} = (1-\lambda)(-2-\lambda) - 4 =$$

$$= -2 - \lambda + 2\lambda + \lambda^2 - 4 =$$

$$= \lambda^2 + \lambda - 6 = (x+3)(x-2)$$

$$(\lambda+3)(\lambda-2) = 0 \quad x = -3 \quad \lambda = 2$$

$\lambda = -3$:

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

setting $x_2 = 1$

$2x_1 + 4x_2 = 0$

$2x_1 + 4 = 0 \quad x_1 = -2$

$$E_{-3} = \text{span}\left\{\begin{bmatrix} -2 \\ 1 \end{bmatrix}\right\}$$

$\lambda = 2$:

$$\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \sim \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$x_2 = 2$

$x_1 = 1$

$$E_2 = \text{span}\left\{\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right\}$$

4.4

Need to compute the following determinant

$$\begin{vmatrix} -\lambda & -1 & 1 & 1 \\ -1 & 1-\lambda & -2 & 3 \\ 2 & -1 & -\lambda & 0 \\ 1 & -1 & 1 & -\lambda \end{vmatrix}$$

expanding across row 3

$$= (-1)^4 (2) \det(A_{3,1}) + (-1)^5 (-1) \det(A_{3,2}) + (-1)^6 (-\lambda) \det(A_{3,3})$$

$$= 2\det(A_{3,1}) + \det(A_{3,2}) - \lambda \det(A_{3,3})$$

$$\det(A_{3,1}) = \begin{vmatrix} -1 & 1 & 1 \\ 1-\lambda & -2 & 3 \\ -1 & 1 & -\lambda \end{vmatrix} = (-1)\begin{vmatrix} -2 & 3 \\ 1 & -\lambda \end{vmatrix} - (1)\begin{vmatrix} 1-\lambda & 3 \\ -1 & -\lambda \end{vmatrix} + (1)\begin{vmatrix} 1-\lambda & -2 \\ -1 & 1 \end{vmatrix}$$

$$= -(2\lambda - 3) - (-\lambda + \lambda^2 + 3) + (1 - \lambda - 2) =$$

$$= -2\lambda + \cancel{3} + \lambda - \lambda^2 - \cancel{3} + 1 - \lambda - 2 = -\lambda^2 - 2\lambda - 1$$

$$-\lambda^2-2\lambda-1 = -(\lambda+1)^2$$

$$\det(A_{3,2}) = \begin{vmatrix} -\lambda & 1 & 1 \\ -1 & -2 & 3 \\ 1 & 1 & -\lambda \end{vmatrix} = (-\lambda)\begin{vmatrix} -2 & 3 \\ 1 & -\lambda \end{vmatrix}$$

$$-(1)\begin{vmatrix} -1 & 3 \\ 1 & -\lambda \end{vmatrix} + (1)\begin{vmatrix} -1 & -2 \\ 1 & 1 \end{vmatrix}$$

$$= (-\lambda)(2\lambda-3) - (\lambda-3) + (-1+2) =$$

$$= -2\lambda^2+3\lambda-\lambda+3+1 = -2\lambda^2+2\lambda+4$$

$$\det(A_{3,3}) = \begin{vmatrix} -\lambda & -1 & 1 \\ -1 & 1-\lambda & 3 \\ 1 & -1 & -\lambda \end{vmatrix} = (-\lambda)\begin{vmatrix} 1-\lambda & 3 \\ -1 & -\lambda \end{vmatrix}$$

$$-(-1)\begin{vmatrix} -1 & 3 \\ 1 & -\lambda \end{vmatrix} + (1)\begin{vmatrix} -1 & 1-\lambda \\ 1 & -1 \end{vmatrix}$$

$$= -\lambda\left((1-\lambda)\cdot(-\lambda)+3\right) + (\lambda-3) + (1-1+\lambda)$$

$$= -\lambda(-\lambda+\lambda^2+3) + (\lambda-3) + \lambda$$

$$= \lambda^2-\lambda^3-3\lambda+\lambda-3+\lambda =$$

$$= -\lambda^3+\lambda^2-\lambda-3$$

$\det(A_{3,4}) = 0$ since $a_{34} = 0$

plugging in $\det(A_{3,1})$, $\det(A_{3,2})$, $\det(A_{3,3})$:

$$\det(A - xI) = -2(x+1)^2 + (-2\lambda^2 + 2\lambda + 4)$$

$$-x(-\lambda^3 + \lambda^2 - \lambda - 3) =$$

$$= -2(x+1)^2 - 2x^2 + 2x + 4 + x^4 - x^3 + x^2 + 3x$$

$$= -2(x+1)^2 + x^4 - x^3 - x^2 + 5x + 4 =$$

$$= -2x^2 - 4x - 2 + x^4 - x^3 - x^2 + 5x + 4 =$$

$$= x^4 - x^3 - 3x^2 + x + 2 = 0$$

Using rational root theorem, the possible root are $\pm 1, \pm 2$

plugging in each in the above gives $x = \pm 1, 2$ as the eigenvalues.

Now we need to find eigenvectors correspondy to the eigenvalues:

$\lambda = -1$

$$\begin{bmatrix} 1 & -1 & 1 & 1 \\ -1 & 2 & -2 & 3 \\ 2 & -1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{bmatrix} \begin{matrix} \\ +R_1 \\ -2R_1 \\ -R_1 \end{matrix} \rightsquigarrow \begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & -1 & 4 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ -R_2 \\ \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 0 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_4 = 0$

$x_2 - x_3 = 0$

$x_2 = x_3 = 1$

$x_1 = 0$

$$E_{-1} = \text{span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$x = 1$:

$$\begin{bmatrix} -1 & -1 & 1 & 1 \\ -1 & 0 & -2 & 3 \\ 2 & -1 & -1 & 0 \\ 1 & -1 & 1 & -1 \end{bmatrix} \begin{matrix} \\ -R_1 \\ +2R_1 \\ +R_1 \end{matrix} \rightarrow \begin{bmatrix} -1 & -1 & 1 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & -3 & 1 & 2 \\ 0 & -2 & 2 & 0 \end{bmatrix} \begin{matrix} \\ \\ +3R_2 \\ +2R_2 \end{matrix}$$

$$\downarrow$$

$$\begin{bmatrix} -1 & -1 & 1 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & -8 & 8 \\ 0 & 0 & -4 & 4 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -1 & 1 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & -8 & 8 \\ 0 & 0 & -4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_4 = 1$

$x_3 = 1$

$x_2 - 1 = 0 \quad x_2 = 1$

$-x_1 + 1 = 0 \quad x_1 = 1$

$$E_1 = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

$x = 2$:

$$\begin{bmatrix} -2 & -1 & 1 & 1 \\ -1 & -1 & -2 & 3 \\ 2 & -1 & -2 & 0 \\ 1 & -1 & 1 & -2 \end{bmatrix} \xrightarrow[\text{swap rows } 1,2]{} \begin{bmatrix} -1 & -1 & -2 & 3 \\ -2 & -1 & 1 & 1 \\ 2 & -1 & -2 & 0 \\ 1 & -1 & 1 & -2 \end{bmatrix} \begin{matrix} -2R_1 \\ +2R_1 \\ +R_1 \\ \\ \end{matrix}$$

$$\downarrow$$

$$\begin{bmatrix} -1 & -1 & -2 & 3 \\ 0 & 1 & 5 & -5 \\ 0 & 0 & 9 & -9 \\ 0 & 0 & 9 & -9 \end{bmatrix} \leftarrow \begin{bmatrix} -1 & -1 & -2 & 3 \\ 0 & 1 & 5 & -5 \\ 0 & -3 & -6 & 6 \\ 0 & -2 & -1 & 1 \end{bmatrix} \begin{matrix} +3R_2 \\ +2R_2 \end{matrix}$$

$x_4 = 1$

$x_3 = 1$

$x_2 = 0$

$x_1 = 1$

$$E_2 = \text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

4.5

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ diagonalizable because it is symmetric

invertible because full rank

$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ diagonalizable because it is symmetric

not invertible because not full rank.

$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$(1-\lambda)^2 = 0$

$\lambda = 1$

$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$E_1 = \text{span}\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$

which means that the dimensionality of the eigenspace is 1, so it is not <u>diagonalizable</u>

Full rank, so invertible.

$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

$\begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = 0$

Not invertible, 0 determinant

Not symmetric, so not diagonalizable

4.6

$$\begin{bmatrix} 2 & 3 & 0 \\ 1 & 4 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{vmatrix} 2-\lambda & 3 & 0 \\ 1 & 4-\lambda & 3 \\ 0 & 0 & 1-\lambda \end{vmatrix} = (2-\lambda)\begin{vmatrix} 4-\lambda & 3 \\ 0 & 1-\lambda \end{vmatrix} - (3)\begin{vmatrix} 1 & 3 \\ 0 & 1-\lambda \end{vmatrix}$$

$$= (2-\lambda)(\lambda^2 - 5\lambda + 4) - 3(1-\lambda) =$$

$$= 2\lambda^2 - 10\lambda + 8 - \lambda^3 + 5\lambda^2 - 4\lambda - 3 + 3\lambda =$$

$$= -\lambda^3 + 7\lambda^2 - 11\lambda + 5$$

possible roots are $\pm 5, \pm 1$

So eigen values are 1 and 5

$\lambda = -5$: $(-5)^3 + 7(25) + 60$

$= -125 + 175 + 60$ ✗

$\lambda = 5$: $-(5)^3 + 7(25) - 50$

$= -125 + 175 - 50$ ✓

$\lambda = 1$: $-1 + 7 - 11 + 5$ ✓

$\lambda = -1$: $1 + 7 + 11 + 5$ ✗

For $\lambda = 5$

$$\begin{bmatrix} -3 & 3 & 0 \\ 1 & -1 & 3 \\ 0 & 0 & -4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_3 = 0$

$x_1 - x_2 = 0$

$x_2 = 1 \quad x_1 = -1$

$$E_5 = \text{span}\left(\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}\right)$$

$\lambda = 1$:

$$\begin{bmatrix} 1 & 3 & 0 \\ 1 & 3 & 3 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_3 = 0$

$x_1 + 3x_2 = 0$

$x_1 = -3 \quad x_2 = 1$

$$E_1 = \text{span}\left(\begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}\right)$$

The geometric multiplicity for $\lambda=1$ is 1, same for $\lambda=5$,
so we do not have enough eigenvectors $\rightarrow$ not diagonalizable

b)

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad \begin{vmatrix} 1-\lambda & 1 & 0 & 0 \\ 0 & -\lambda & 0 & 0 \\ 0 & 0 & -\lambda & 0 \\ 0 & 0 & 0 & -\lambda \end{vmatrix}$$

$$= -\lambda \det(A_{4,4})$$

$$\det(A_{4,4}) = \begin{vmatrix} 1-\lambda & 1 & 0 \\ 0 & -\lambda & 0 \\ 0 & 0 & -\lambda \end{vmatrix} = -\lambda \begin{vmatrix} 1-\lambda & 1 \\ 0 & -\lambda \end{vmatrix} = -\lambda(-\lambda+\lambda^2)$$

$$= \lambda^2 - \lambda^3$$

$$-\lambda(\lambda^2-\lambda^3) = -\lambda^3+\lambda^4 = \lambda^3(\lambda-1)$$

$$\lambda^3(\lambda-1)=0 \qquad \lambda=0 \quad \lambda=1$$

$\lambda=0$:

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$E_0 = \text{span}\left(\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 0 \\ 0 \end{bmatrix}\right)$
which has dimension 3,
i.e geometric multiplicity is 3

$\lambda=1$:

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_4=0$
$x_3=0$
$x_2=0$
$x_1=1$

$E_1 = \text{span}\left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}\right)$
which has dimension 1
and geometric multiplicity of 1

So we have 4 linearly independent eigenvectors, which means the matrix is diagonalizable.

4.7

a) $\begin{vmatrix} -\lambda & 1 \\ -8 & 4-\lambda \end{vmatrix} = -\lambda(4-\lambda) + 8 = \lambda^2 - 4\lambda + 8$

using quadratic formula:

$\lambda = \dfrac{4 \pm \sqrt{16-32}}{2}$ $\qquad \lambda = \dfrac{4 \pm 4i}{2} = 2 \pm 2i$

Yes it is diagonalizable but no real eigenvalues

$\lambda = 2 + 2i$

$$\begin{bmatrix} -2-2i & 1 \\ -8 & 2-2i \end{bmatrix} \xrightarrow[-(2-2i)R_1]{} \begin{bmatrix} -2-2i & 1 \\ 0 & 0 \end{bmatrix}$$

$(-2-2i)(-2-2i) = 4 + \cancel{4}i - \cancel{4}i + 4i^2 = 8$

$(-2-2i)x_1 + x_2 = 0$

$-2(1+i)x_1 = -x_2$

$2(1+i)x_1 = x_2$

$x_1 = 1 \qquad x_2 = 2(1+i)$

$E_{2+2i} = \text{span}\left(\begin{bmatrix} 1 \\ 2+2i \end{bmatrix}\right)$

$x = 2 - 2i$

$$\begin{bmatrix} -2+2i & 1 \\ -8 & 2+2i \end{bmatrix}_{-R_1(2+2i)} \rightarrow \begin{bmatrix} -2+2i & 1 \\ 0 & 0 \end{bmatrix}$$

$(-2+2i)(2+2i) = -4 - 4i + 4i + 4i^2 = -8$ ✓

$(-2+2i)x_1 + x_2 = 0$

$x_1 = 1$

$x_2 = 2 - 2i$

$E_{2-2i} = \text{span}\left(\begin{bmatrix} 1 \\ 2-2i \end{bmatrix}\right)$

(b) symmetric, so diagonalizable.

$$\begin{vmatrix} 1-\lambda & 1 & 1 \\ 1 & 1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{vmatrix} = (1-\lambda)\begin{vmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} - \begin{vmatrix} 1 & 1 \\ 1 & 1-\lambda \end{vmatrix} + \begin{vmatrix} 1 & 1-\lambda \\ 1 & 1 \end{vmatrix}$$

$$= (1-\lambda)(\cancel{1}-\lambda-\lambda+\lambda^2-\cancel{1}) - (\cancel{1}-\lambda-\cancel{1}) + (1-1+\lambda)$$

$$= (1-\lambda)(\lambda^2-2\lambda) + \lambda + \lambda = \lambda^2 - \cancel{2\lambda} - \lambda^3 + 2\lambda^2 + \cancel{2\lambda}$$

$$= -\lambda^3 + 3\lambda^2.$$

$$\lambda^2(-\lambda+3) = 0$$

$\lambda = 0 \quad \lambda = 3$

$\lambda = 3$:

$$\begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \xrightarrow{\text{swap rows } 1,2} \begin{bmatrix} 1 & -2 & 1 \\ -2 & 1 & 1 \\ 1 & 1 & -2 \end{bmatrix} \begin{matrix} \\ +2R_1 \\ -R_1 \end{matrix} \rightarrow \begin{bmatrix} 1 & -2 & 1 \\ 0 & -3 & 3 \\ 0 & 3 & -3 \end{bmatrix} \begin{matrix} \\ \\ -R_2 \end{matrix} \rightarrow \begin{bmatrix} 1 & -2 & 1 \\ 0 & -3 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & -3 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-3x_2 + 3x_3 = 0$

$x_2 = 1 \quad x_3 = 1$

$x_1 - 2 + 1 = 0$

$x_1 - 1 = 0$

$x_1 = 1$

$$E_3 = \text{span}\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right)$$

$\lambda = 0$:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{matrix} \\ -R_1 \\ -R_1 \end{matrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

x_2 and x_3 are free variables

$x_1 + x_2 + x_3 = 0$

$x_1 = -x_2 - x_3$

$x_3 = 0 \quad x_2 = 1 \quad x_1 = -1$

$x_3 = 1 \quad x_2 = 0 \quad x_1 = -1$

$E_0 = \text{span}\left(\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right)$

$A = PDP^{-1}$

$P = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

note that the found eigenvectors are not orthonormal

$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

c)

$$\begin{vmatrix} 5-\lambda & 4 & 2 & 1 \\ 0 & 1-\lambda & -1 & -1 \\ -1 & -1 & 3-\lambda & 0 \\ 1 & 1 & -1 & 2-\lambda \end{vmatrix} = (-1)^{3+1}(-1)\det(A_{3,1}) + (-1)^{3+2}(-1)\det(A_{3,2}) + (-1)^{3+3}(3-\lambda)\det(A_{3,3})$$

expand along row 3

$$= -\det(A_{3,1}) + \det(A_{3,2}) + (3-\lambda)\det(A_{3,3})$$

$$\det(A_{3,1}) = \begin{vmatrix} 4 & 2 & 1 \\ 1-\lambda & -1 & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix}$$

$$= 4\begin{vmatrix} -1 & -1 \\ -1 & 2-\lambda \end{vmatrix} - 2\begin{vmatrix} 1-\lambda & -1 \\ 1 & 2-\lambda \end{vmatrix} + \begin{vmatrix} 1-\lambda & -1 \\ 1 & -1 \end{vmatrix}$$

$$= 4(\lambda - 2 - 1) - 2(2 - 3\lambda + \lambda^2 + 1) + (\lambda - 1 + 1)$$

$$= 4\lambda - 12 - 2\lambda^2 + 6\lambda - 6 + \lambda$$

$$= -2\lambda^2 + 11\lambda - 18$$

$$\det(A_{3,2}) = \begin{vmatrix} 5-\lambda & 2 & 1 \\ 0 & -1 & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix} =$$

$$= (5-\lambda)\begin{vmatrix} -1 & -1 \\ -1 & 2-\lambda \end{vmatrix} - 2\begin{vmatrix} 0 & -1 \\ 1 & 2-\lambda \end{vmatrix} + \begin{vmatrix} 0 & -1 \\ 1 & -1 \end{vmatrix}$$

$$= (5-\lambda)(\overset{(\lambda-3)}{\lambda-2-1}) - 2(1) + (1)$$

$$= 5\lambda - 15 - \lambda^2 + 3\lambda - 1 = -\lambda^2 + 8\lambda - 16$$

$$A_{3,3} = \begin{vmatrix} 5-\lambda & 4 & 1 \\ 0 & 1-\lambda & -1 \\ 1 & 1 & 2-\lambda \end{vmatrix} =$$

$$= (5-\lambda)\begin{vmatrix} 1-\lambda & -1 \\ 1 & 2-\lambda \end{vmatrix} - 4\begin{vmatrix} 0 & -1 \\ 1 & 2-\lambda \end{vmatrix} + \begin{vmatrix} 0 & 1-\lambda \\ 1 & 1 \end{vmatrix}$$

$$= (5-\lambda)(\overset{(\lambda^2-3\lambda+2)}{2-2\lambda-\lambda+\lambda^2+1}) - 4(1) + (-1+\lambda)$$

$$= 5\lambda^2 - 15\lambda + 15 - \lambda^3 + 3\lambda^2 - 3\lambda - 4 - 1 + \lambda$$

$$= -\lambda^3 + 8\lambda^2 - 17\lambda + 10$$

$$\det(A-\lambda I) = -(-2\lambda^2 + 11\lambda - 18) + (-\lambda^2 + 8\lambda - 16)$$
$$+ (3-\lambda)(-\lambda^3 + 8\lambda^2 - 17\lambda + 10)$$

$$= 2\lambda^2 - 11\lambda + 18 - \lambda^2 + 8\lambda - 16 + \dots$$
$$= \lambda^2 - 3\lambda + 2 + (-3\lambda^3 + 24\lambda^2 - 51\lambda + 30 + \lambda^4 - 8\lambda^3 + 17\lambda^2 - 10\lambda)$$

$$= \lambda^4 - 11\lambda^3 + 42\lambda^2 - 64\lambda + 32$$

using rational root theorem, possible roots are $\pm 1, \pm 2, \pm 4, \pm 8, \pm 16, \pm 32$

Trying these out, we get $\lambda = 1, 2, 4$

$\lambda = 1$

$$\begin{bmatrix} 4 & 4 & 2 & 1 \\ 0 & 0 & -1 & -1 \\ -1 & -1 & 2 & 0 \\ 1 & 1 & -1 & 1 \end{bmatrix} \xrightarrow[R_1, R_4]{\text{swap}} \begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & 0 & -1 & -1 \\ -1 & -1 & 2 & 0 \\ 4 & 4 & 2 & 1 \end{bmatrix} \begin{matrix} \\ \\ +R_1 \\ -4R_1 \end{matrix} \longrightarrow \begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 6 & -3 \end{bmatrix} \begin{matrix} \\ \\ +R_2 \\ +6R_2 \end{matrix}$$

$$\downarrow$$

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & -9 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xleftarrow[R_{3,4}]{\text{swap}} \begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -9 \end{bmatrix}$$

$x_3 = 0$

$x_2 = 0$

$x_1 + x_2 = 0$

$x_1 = -x_2$

$$E_1 = \text{span}\left(\begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}\right)$$

$\lambda = 2$:

$$\begin{bmatrix} 3 & 4 & 2 & 1 \\ 0 & -1 & -1 & -1 \\ -1 & -1 & 1 & 0 \\ 1 & 1 & -1 & 0 \end{bmatrix} \xrightarrow{\text{swap } R_{1,4}} \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & -1 & -1 & -1 \\ -1 & -1 & 1 & 0 \\ 3 & 4 & 2 & 1 \end{bmatrix} \begin{matrix} \\ \\ +R_1 \\ -3R_1 \end{matrix} \rightarrow \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 5 & 1 \end{bmatrix} \begin{matrix} \\ \\ \\ +R_2 \end{matrix}$$

$$\downarrow$$

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 \end{bmatrix} \xrightarrow{\text{swap } R_{3,4}} \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & -1 & -1 & -1 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_4 = 1$$
$$x_3 = 0$$
$$-x_2 - 1 = 0$$
$$x_2 = -1$$
$$x_1 = 1$$

$$E_2 = \text{span}\left(\begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}\right)$$

$\lambda = 4$:

$$\begin{bmatrix} 1 & 4 & 2 & 1 \\ 0 & -3 & -1 & -1 \\ -1 & -1 & -1 & 0 \\ 1 & 1 & -1 & -2 \end{bmatrix} \xrightarrow{\text{swap } R_{1,4}} \begin{bmatrix} 1 & 1 & -1 & -2 \\ 0 & -3 & -1 & -1 \\ -1 & -1 & -1 & 0 \\ 1 & 4 & 2 & 1 \end{bmatrix} \begin{matrix} \\ \\ +R_1 \\ -R_1 \end{matrix}$$

$$\downarrow$$

$$\begin{bmatrix} 1 & 1 & -1 & -2 \\ 0 & -3 & -1 & -1 \\ 0 & 0 & -2 & -2 \\ 0 & 3 & 3 & 3 \end{bmatrix} \begin{matrix} \\ \\ \\ +R_2 \end{matrix} \leftarrow \begin{bmatrix} 1 & 1 & -1 & -2 \\ 0 & -3 & -1 & -1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 2 & 2 \end{bmatrix} \begin{matrix} \\ \\ \\ +R_3 \end{matrix}$$

$$\begin{bmatrix} 1 & 1 & -1 & -2 \\ 0 & -3 & -1 & -1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_4 = 1$

$-x_3 - x_4 = 0$

$-x_3 = x_4$

$x_3 = -1$

$-3x_2 + 1 - 1 = 0 \quad x_2 = 0$

$x_1 + 1 - 2 = 0$

$x_1 - 1 = 0$

$x_1 = 1$

$$E_4 = \text{span}\left(\begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}\right)$$

We have 3 eigenvectors, so we cannot form a basis for $\mathbb{R}^4$. Not diagonalizable.

d)

$$\begin{vmatrix} 5-\lambda & -6 & -6 \\ -1 & 4-\lambda & 2 \\ 3 & -6 & -4-\lambda \end{vmatrix} = (5-\lambda)\begin{vmatrix} 4-\lambda & 2 \\ -6 & -4-\lambda \end{vmatrix}$$

$$+ 6\begin{vmatrix} -1 & 2 \\ 3 & -4-\lambda \end{vmatrix} - 6\begin{vmatrix} -1 & 4-\lambda \\ 3 & -6 \end{vmatrix}$$

$$= (5-\lambda)\left(-16 + \overbrace{\cancel{4\lambda} - \cancel{4\lambda} + \lambda^2}^{\lambda^2-4} + 12\right) + 6\left(\overbrace{4+\lambda-6}^{\lambda-2}\right) - 6\left(\overbrace{6-12+3\lambda}^{3\lambda-6}\right)$$

$$= 5\lambda^2 - 20 - \lambda^3 + \underline{4\lambda} + \underline{6\lambda} - 12 - \underline{18\lambda} + 36$$

$= -\lambda^3 + 5\lambda^2 - 8\lambda + 4$

roots are $\lambda = 1, 2$

$\lambda = 1$:

$$\begin{bmatrix} 4 & -6 & -6 \\ -1 & 3 & 2 \\ 3 & -6 & -5 \end{bmatrix} \xrightarrow[R_{1,2}]{\text{swap}} \begin{bmatrix} -1 & 3 & 2 \\ 4 & -6 & -6 \\ 3 & -6 & -5 \end{bmatrix} \begin{matrix} \\ +4R_1 \\ +3R_1 \end{matrix} \rightarrow \begin{bmatrix} -1 & 3 & 2 \\ 0 & 6 & 2 \\ 0 & 3 & 1 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} -1 & 3 & 2 \\ 0 & 3 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$x_3 = 3$

$3x_2 + 3 = 0$

$x_2 = -1$

$-x_1 - 3 + 6 = 0$

$x_1 = 3$

$$E_1 = \text{span}\left(\begin{bmatrix} 3 \\ -1 \\ 3 \end{bmatrix}\right)$$

$\lambda = 2$:

$$\begin{bmatrix} 3 & -6 & -6 \\ -1 & 2 & 2 \\ 3 & -6 & -6 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_3 = 0$

$x_2 = 1$

$x_1 = 2$

$x_3 = 1$

$x_2 = 0$

$x_1 = 2$

$$E_2 = \text{span}\left(\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}\right)$$

we have 3 eigenvectors, so can form a basis:

$$P = \begin{bmatrix} 2 & 2 & 3 \\ 1 & 0 & -1 \\ 0 & 1 & 3 \end{bmatrix} \qquad D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

1.8

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} = \begin{bmatrix} 13 & 12 & 2 \\ 12 & 13 & -2 \\ 2 & -2 & 8 \end{bmatrix}$$

Now do eigenvalue decomposition:

$$\det(A^T A - \lambda I) = \begin{vmatrix} 13-\lambda & 12 & 2 \\ 12 & 13-\lambda & -2 \\ 2 & -2 & 8-\lambda \end{vmatrix}$$

$$= (13-\lambda)\begin{vmatrix} 13-\lambda & -2 \\ -2 & 8-\lambda \end{vmatrix} - 12\begin{vmatrix} 12 & -2 \\ 2 & 8-\lambda \end{vmatrix} + 2\begin{vmatrix} 12 & 13-\lambda \\ 2 & -2 \end{vmatrix}$$

$$= (13-\lambda)\overbrace{(104 - 8\lambda - 13\lambda + \lambda^2 - 4)}^{(100-21\lambda+\lambda^2)} - 12\overbrace{(96 - 12\lambda + 4)}^{(100-12\lambda)} + 2\overbrace{(-24 - 26 + 2\lambda)}^{-50}$$

$$= 1300 - 373\lambda + 13\lambda^2 - 100\lambda + 21\lambda^2 - \lambda^3 - 1200 + 144\lambda - 100 + 4\lambda$$

$$= -\lambda^3 + 34\lambda^2 - 225\lambda = -\lambda(\lambda^2 - 34\lambda + 225)$$

$$\frac{34 \pm \sqrt{(34)^2 - 4 \cdot 225}}{2} = \frac{34 \pm 16}{2}$$

$$\begin{array}{r} 34 \cdot 34 \\ \hline 136 \\ 102 \\ \hline 1156 \end{array} \qquad 16^2 = 256$$

$1156 - 900 = 256$

roots are $\lambda = 0$, $\lambda = 25$, $\lambda = 9$

Singular values are $0, 3, 5$

$$\begin{bmatrix} 13 & 12 & 2 \\ 12 & 13 & -2 \\ 2 & -2 & 8 \end{bmatrix} \begin{matrix} -R_2 \\ \\ \\ \end{matrix} \rightarrow \begin{bmatrix} 1 & -1 & 4 \\ 12 & 13 & -2 \\ 2 & -2 & 8 \end{bmatrix} \begin{matrix} \\ -12R_1 \\ -2R_1 \end{matrix} \rightarrow \begin{bmatrix} 1 & -1 & 4 \\ 0 & 25 & -50 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 1 & -1 & 4 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 4 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_3 = 1$ $\quad x_2 - 2 = 0$ $\quad x_1 - 2 + 4 = 0$

$x_2 = 2$ $\quad x_1 = -2$

$$E_0 = \text{span}\left(\begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}\right)$$

normalized $= \frac{1}{3}\begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$

$\lambda = 9$:

$$\begin{bmatrix} 4 & 12 & 2 \\ 12 & 4 & -2 \\ 2 & -2 & -1 \end{bmatrix} \begin{matrix} \\ -3R_1 \\ -\frac{1}{2}R_1 \end{matrix} \rightarrow \begin{bmatrix} 4 & 12 & 2 \\ 0 & -32 & -8 \\ 0 & -8 & -2 \end{bmatrix} \begin{matrix} \\ /-8 \\ /-2 \end{matrix} \rightarrow \begin{bmatrix} 4 & 12 & 2 \\ 0 & 4 & 1 \\ 0 & 4 & 1 \end{bmatrix}$$

$x_3 = 4$

$4x_3 + 4 = 0 \quad x_3 = -1$

$4x_1 - 12 + 8 = 0 \quad 4x_1 - 4 = 0 \quad x_1 = 1$

$$E_9 = \text{span}\left(\begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}\right)$$

$1^2 + (-1)^2 + (4)^2 = 18 \quad \sqrt{18} = 3\sqrt{2}$

normalized $\frac{1}{3\sqrt{2}}\begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$

$\lambda = 25$:

$$\begin{bmatrix} -12 & 12 & 2 \\ 12 & -12 & -2 \\ 2 & -2 & -17 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -2 & -17 \\ -12 & 12 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ +6R_1 \\ \\ \end{matrix} \rightarrow \begin{bmatrix} 2 & -2 & -17 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{bmatrix}$$

$x_3 = 0$

$2x_1 - 2x_2 = 0$

$x_1 = 1 \quad x_2 = 1$

$$E_{25} = \text{span}\left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right)$$

normalized $= \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

So we have $\Sigma = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix}$

$$AA^T = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$$

Check that we get the same eigenvalues:

$$\det(AA^T - \lambda I) = \begin{vmatrix} 17-\lambda & 8 \\ 8 & 17-\lambda \end{vmatrix} = 289 - 17\lambda - 17\lambda + \lambda^2 - 64$$

$$= 289 - 34\lambda + \lambda^2 - 64$$

$$= \lambda^2 - 34\lambda + 225$$

$$\frac{34 \pm \sqrt{34^2 - 4 \cdot 225}}{2} = \frac{34 \pm 16}{2}$$

$$\begin{array}{r} 34 \\ \times 34 \\ \hline 136 \\ +102 \\ \hline 1156 \end{array}$$

$1156 - 900 = 256$

$\lambda = 25, \lambda = 9$ ✓ same as before

$\lambda = 25$:

$$\begin{bmatrix} -8 & 8 \\ 8 & -8 \end{bmatrix} \to \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$x_2 = 1$
$x_1 = 1$

$E_{25} = \text{span}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$

normalized $= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\lambda = 9$:

$$\begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} \to \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$x_1 = 1$
$x_2 = -1$

$E_9 = \text{span}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right)$

normalized $= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\sigma_1 = 25$ $\quad u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{5\sqrt{2}} \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{5\sqrt{2}} \begin{bmatrix} 5 \\ 5 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ✓

$\sigma_2 = 9$ $\quad u_2 = \frac{1}{\sigma_2} A v_2 = \frac{1}{9\sqrt{2}} \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} = \frac{1}{9\sqrt{2}} \begin{bmatrix} 9 \\ -9 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ ✓

So we get $A = U\Sigma V^T = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix}\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{3\sqrt{2}} & \frac{-1}{3\sqrt{2}} & \frac{4}{3\sqrt{2}} \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$

4.9

$A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$ $\quad A^TA = \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$

$\det(A - \lambda I) = \begin{vmatrix} 5-\lambda & 3 \\ 3 & 5-\lambda \end{vmatrix} = 25 - 5\lambda - 5\lambda + \lambda^2 - 9 =$

$= \lambda^2 - 10\lambda + 16$

$(\lambda - 8)(\lambda - 2)$

$\lambda = 8 \quad \lambda = 2$

$\lambda = 2$:

$\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$E_2 = \text{span}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right)$

normalized $v_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\lambda = 8$:

$\begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$x_1 = 1$
$x_2 = 1$

normalized $v_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$E_8 = \text{span}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$

$\sigma_1 = \sqrt{8} = 2\sqrt{2}$ $\quad \frac{1}{\sigma_1} Av_1 = \frac{1}{2\sqrt{2}}\begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\sigma_2 = \sqrt{2}$ $\quad \frac{1}{\sigma_2} Av_2 = \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$A = U\Sigma V^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix}\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$

4.10

Same matrix as 4.8

$$A = U\Sigma V^T = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix}\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{3\sqrt{2}} & \frac{-1}{3\sqrt{2}} & \frac{4}{3\sqrt{2}} \\ \frac{-2}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

$$A_1 = \sigma_1 u_1 v_1^T = 5\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix} = 5\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

4.11

$A^TAx = \lambda x$ where λ is an eigenvalue of A^TA

$AA^T(Ax) = \lambda(Ax)$ we see that λ is also an eigenvalue of AA^T

$AA^Ty = \lambda y$, $y = Ax$

4.12

$$Ax = \overset{m\times m}{U}\,\overset{m\times n}{\Sigma}\,\overset{n\times n}{V^T}\,\overset{n\times 1}{x}$$

$$= U\Sigma V^T V c$$

$$= U\Sigma c$$

$x = x_1e_1 + x_2e_2 + \dots + x_ne_n$

but we can also express x in the basis $\{\vec{v}_1, \dots, \vec{v}_n\}$

$x = c_1v_1 + c_2v_2 + \dots + c_nv_n$

$$\|Ax\|_2^2 = (Ax)\cdot(Ax) = (Ax)^T(Ax) = (U\Sigma c)^T(U\Sigma c) =$$

$$= c^T\Sigma^T \cancel{U^T}\cancel{U}\Sigma c = c^T\Sigma^T\Sigma c = \overset{1\times n}{c^T}\overset{n\times n}{\begin{bmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_n^2 \end{bmatrix}} c =$$

$$= \sum_{i=1}^{n} \sigma_i^2 c_i^2$$

$$\|x\|_2^2 = x^Tx = \sum_{i=1}^{n} c_i v_i^T \cdot \sum_{j=1}^{n} c_j v_j = \sum_{i=1}^{n}\sum_{j=1}^{n} c_ic_j \underbrace{v_i^T\cdot v_j}_{\substack{1 \text{ if } i=j \\ \text{else } 0}} = \sum_{i=1}^{n} c_i^2$$

$$\frac{\|Ax\|_2^2}{\|x\|_2^2} = \frac{\sum_{i=1}^{n} \sigma_i^2 c_i^2}{\sum_{i=1}^{n} c_i^2} \leq \frac{\sum_{i=1}^{n} \sigma_1^2 c_i^2}{\sum_{i=1}^{n} c_i^2} = \frac{\sigma_1^2 \cancel{\sum_{i=1}^{n} c_i^2}}{\cancel{\sum_{n=1}^{n} c_i^2}} = \sigma_1^2$$

plug in σ_1

where σ_1 is the largest singular value in Σ.

Taking the square root:

$$\frac{\|Ax\|_2}{\|x\|_2} = \frac{\sqrt{\sum_{i=1}^{n} \sigma_i^2 c_i^2}}{\sqrt{\sum_{i=1}^{n} c_i^2}} \leq \sigma_1$$

So we have

$$\|A\|_2 = \max_x \frac{\|Ax\|_2}{\|x\|} \leq \sigma_1$$

We can set $\vec{x} = 1\vec{v}_1 + 0\vec{v}_2 + \dots + 0\vec{v}_n$

$Av_1 = U\Sigma V^T v_1 = \sigma_1 \vec{u}_1$, so $\|Av_1\|_2 = \sqrt{\sigma_1^2 \underbrace{u_1^T u_1}_{=1}} = \sqrt{\sigma_1^2} = \sigma_1$

also note $\|\vec{u}_m\| = 1$ and $\|\vec{v}_n\| = 1$

We can then write the following inequality:

$$\sigma_1 = \frac{\|Av_1\|_2}{\|v_1\|_2} \leq \max_x \frac{\|Ax\|_2}{\|x\|_2}$$

But before we had $\max_x \frac{\|Ax\|_2}{\|x\|_2} \leq \sigma_1$, so the only way these two inequalities can hold is if $\frac{\|Ax\|_2}{\|x\|_2} = \sigma_1$

from which we get $\|A\|_2 = \sigma_1$